Lecture 9: Applied Regression in Economics (Part II)

Interpretation, Diagnostics, and Credibility

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Where we are in the course

- We can now: collect data (APIs), clean/merge, visualize, and write notebooks as reports.
- This week: **Applied Regression in Economics** (OLS as a workhorse).
- Next: **Causal inference** (prediction vs. causation, DiD intuition).

Learning objectives

By the end of today, you should be able to:

1. Interpret OLS coefficients *in context* (units, magnitudes, elasticities).
2. Choose functional forms (level-level, log-level, log-log) intentionally.
3. Explain omitted variable bias (OVB) intuitively (no proofs).
4. Recognize heteroskedasticity and use robust standard errors.
5. Run basic diagnostics: residual patterns, outliers, influence.

OLS refresher

Model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

OLS chooses $\hat{\beta}$ to minimize $\sum_i \hat{u}_i^2$.

In applied work, the hard part is usually **not** running OLS. It is:

- choosing variables and units,
- interpreting magnitudes,
- evaluating credibility.

Interpretation: always start with units

If you estimate:

$$\mathit{EmploymentRate} = \beta_0 + \beta_1 \, \mathit{MinimumWage} + u$$

then β_1 is "change in employment rate" per \$1 change in minimum wage.

But only if:

- EmploymentRate is in % points (0–100) vs. a share (0–1),
- MinimumWage is in dollars (nominal) vs. real,
- the sample/time period is clearly stated.

Functional forms you will use constantly

1) Level–Level

$$Y = \beta_0 + \beta_1 X + u$$

Interpretation: β_1 is a *unit change* in Y per 1 unit of X.

2) Log–Level

$$\ln(Y) = \beta_0 + \beta_1 X + u$$

Interpretation: $\beta_1 \approx$ % change in Y for a 1-unit change in X.

3) Log–Log

$$\ln(Y) = \beta_0 + \beta_1 \ln(X) + u$$

Interpretation: β_1 is an elasticity.

Example: log interpretation (rule of thumb)

If $\ln(Y)$ is the dependent variable:

$$\ln(Y) = \beta_0 + \beta_1 X + u$$

A 1-unit increase in X changes Y by about $100 \cdot \beta_1$ percent (for small β_1).

More precise:

$$\%\Delta Y \approx 100 \cdot (e^{\beta_1} - 1)$$

Omitted Variable Bias (intuitive)

Suppose the true model is:

$$Wage = \beta_0 + \beta_1 Education + \beta_2 Ability + u$$

But you estimate:

$$Wage = \alpha_0 + \alpha_1 Education + v$$

If **Ability** affects wages and is correlated with education, then α_1 is biased.

Key takeaway: **Regression will attribute omitted effects to included variables.**

Heteroskedasticity: why robust SE

OLS coefficients can still be unbiased under heteroskedasticity, but standard errors can be wrong.

Economic data often have non-constant variance:

- income/wages,
- firm size,
- county populations,
- spending/wealth.

Default in applied work: report robust standard errors.

Python: robust SE in statsmodels

```
import statsmodels.api as sm

X = sm.add_constant(df[["educ", "exper"]])
y = df["wage"]

ols = sm.OLS(y, X).fit()
rob = sm.OLS(y, X).fit(cov_type="HC1") # robust

print(ols.summary())
print(rob.summary())
```

Same coefficients, potentially different standard errors and p-values.

Diagnostics (practical checklist)

- **Residual plot:** patterns suggest misspecification (nonlinearity).
- **Outliers:** large residuals can dominate the fit.
- **Influence:** points with high leverage can shift coefficients.
- **Fit:** R^2 is not a measure of causal credibility.

Goal: build the habit of **checking**, not just estimating.

Prediction vs. causation (teaser for next unit)

A statistically significant coefficient does *not* automatically imply causation.

Today:

- Interpret what the regression is saying.
- Diagnose problems and report uncertainty correctly.

Next:

- Identification and research design (DiD, IV intuition, threats).

In-class workflow today

1. Load dataset (included in notebook; no API keys needed).
2. Run OLS in three forms: level-level, log-level, log-log.
3. Compare conventional vs robust SE.
4. Demonstrate OVB by adding a confounder.
5. Run quick diagnostics: residuals and influence.
6. Write a 3-sentence executive summary (practitioner audience).

Deliverable (end of class)

- Your notebook should include:
  - at least 2 regressions with interpretation,
  - robust SE for your preferred model,
  - one diagnostic plot,
  - a short written summary.